

1. Administrative & Study Identification

Full Study Title: A Translational Study for Prediction of Biomarkers and Identification of Phenotype and Endotype of COPD and Early COPD Outcomes in Chinese Population **Short Title/ Acronym:** COPD Biomarker and Endotype Translational Study **Protocol Number:** D2287R00187 **NCT Number:** NCT06724848 **Sponsor Name:** AstraZeneca **Investigational Product:** N/A (Observational Study) **Phase of Development:** N/A (Observational) **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To describe patient clinical characteristics and treatment patterns, and to identify phenotypes and endotypes associated with differential outcomes in early COPD and moderate-to-very-severe COPD in a Chinese population. **Secondary Objectives:** To evaluate the trajectories of lung function decline, clinical outcomes, and understand the inflammatory and immunological mechanisms driving early COPD and physician-diagnosed COPD.

2.2 Background & Rationale To provide a comprehensive understanding of Chronic Obstructive Pulmonary Disease (COPD) disease progression. Early recognition of COPD is difficult, and the disease is progressive. This translational study aims to identify distinct patient phenotypes and endotypes using biomarkers, which may support the future development of personalized and targeted treatment strategies for the Chinese population.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Observational **Control Type:** Healthy Controls **Blinding:** Open-label (N/A for observational) **Randomization:** N/A

3.2 Planned Enrollment & Duration **Target \$N\$:** Approximately 700 participants total (~450 with moderate-to-very-severe COPD, ~200 with early COPD, and ~50 healthy controls) **Number of Sites:** Multiple outpatient centers and sites in China **Estimated Study Start:** 05-Jun-2024 **Estimated Primary Completion:** 21-Oct-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Males and females aged 30 years or older (50+ for the moderate-to-severe COPD cohort; 30-45 for the early COPD cohort; 30+ for healthy controls). 2. **Moderate-to-severe COPD:** Post-bronchodilation FEV1/FVC < 70%, FEV1 $\geq 25\%$ and <80% of predicted, with presence of respiratory symptoms equivalent to CAT ≥ 10 or mMRC ≥ 2 . 3. **Early COPD:** Smokers or ex-smokers with normal lung function (FEV1/FVC > 70% and FEV1 > 80% of predicted) but demonstrating symptoms (chronic cough, sputum, or breathlessness equivalent to mMRC Grade 1 or higher) and structural risk factors on CT.

4.2 Exclusion Criteria 1. Clinically important pulmonary disease other than COPD (e.g., active lung infection, clinically significant bronchiectasis, pulmonary fibrosis, cystic fibrosis). 2. COPD exacerbation within 2 weeks prior to enrollment that was treated with systemic corticosteroids and/or antibiotics, or led to hospitalization. 3. History of partial or total lung resection, or lung volume reduction surgery within 6 months prior to enrollment. 4. Unstable disorders including severe autoimmune disease, diabetes, thyroid disease, or significant cardio-renal disease (e.g., atrial fibrillation, hypertrophic cardiomyopathy).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: N/A (Observational cohort study; no investigational drug is administered). **Route of Administration:** N/A **Duration of Treatment:** Observational follow-up for an initial 1-year period, which may extend up to 3 years (up to 177 weeks) depending upon emerging data.

5.2 Concomitant & Prohibited Therapy Permitted: Standard of care for COPD (e.g., inhaled corticosteroids, LABA, LAMA) according to the treating physician's discretion. **Prohibited:** N/A (No therapies are strictly prohibited as this is an observational real-world study).

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Spirometry, Questionnaires (CAT, mMRC)
Baseline	Day 0	Enrollment, Biomarker sample collection (blood/sputum), CT scan/Oscillometry
Follow-Up	Months 6, 12	Clinical outcomes assessment, exacerbation tracking, lung function assessment
Long-Term	Years 2, 3	Trajectory of lung function decline tracking (if study is extended)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Identification of patient phenotypes and endotypes associated with differential outcomes in early COPD and moderate-to-severe COPD, predicted by specific biomarker expressions. **Measurement Method:** Central Laboratory Analysis of biomarkers correlated with clinical lung function readouts (Spirometry).

7.2 Secondary & Exploratory Endpoints * Trajectories of lung function decline (measured by annual change in FEV1). * Evaluation of inflammatory and immunological mechanisms on early COPD. * Rates of moderate to severe COPD exacerbations and clinical outcomes on physician-diagnosed COPD.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection of AEs related to study-specific procedures (e.g., phlebotomy). Disease-related events such as acute exacerbations are tracked as primary/secondary clinical outcomes. **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Internal safety and data monitoring mechanisms managed by AstraZeneca.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Enrolled Set (all participants who signed informed consent and met eligibility criteria, \$N \approx 700\$). **Sample Size Justification:** The sample size of ~700 participants is appropriately powered to stratify patients into subgroups based on biomarker profiles and to correlate these with clinical trajectories and endotypes. **Handling of Missing Data:** Mixed-effects models for repeated measures (MMRM) and Multiple Imputation (MI) where applicable for continuous laboratory and spirometry endpoints.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Periodic site monitoring to ensure data quality, sample integrity, and adherence to the observational protocol. **Audit Trail:** eCRF locking and data clarification forms via standard quality audit mechanisms.

11. Study Identifiers & Governance

Primary ID: D2287R00187 **Registry Number:** NCT06724848 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** A Translational Study for Prediction of Biomarkers and Identification of Phenotype and Endotype of COPD and Early COPD Outcomes in Chinese Population
Official Regulatory Title: A Translational Study for Prediction of Biomarkers and Identification of Phenotype and Endotype of COPD and Early COPD Outcomes in Chinese Population

12. Clinical Framework (COPD Specific)

Primary Condition: Chronic Obstructive Pulmonary Disease (COPD) and Early COPD **Diagnosis Threshold:** Post-bronchodilation FEV1/FVC < 70% (Moderate-to-Severe) and FEV1/FVC > 70% with chronic symptoms (Early COPD) **Medical Methodology:** Standard of Care & Observational Biomarker/Endotype Tracking **Optimization Target:** Identification of actionable phenotypes to support personalized treatment strategies

13. Study Procedures & Methodology

Management Pathway: Standard clinical pathway for COPD maintenance and exacerbation management **Education Protocol (HCP):** N/A (Observational standard-of-care protocol) **Education Protocol (Patient):** N/A (Observational standard-of-care protocol) **Quality Audit Mechanism:** Data management review and periodic monitoring of eCRF entries and biological sample handling

14. Observation & Follow-Up Schedule

Total Duration: 52 weeks initially, with a potential extension up to 156-177 weeks (1 to 3 years) **Onsite Visit Cadence:** Baseline, Month 6, Year 1, Year 2, Year 3 **Remote Monitoring Frequency:** As per observational protocol for exacerbation tracking **Checklist Review Interval:** Routine outcome assessments during scheduled observational follow-ups

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				